

DINH CONG THANH

Senior AI Workflow Engineer | Full-stack Developer | AI Automation Engineer | Production RAG & Orchestration

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Showcase **URL:** <https://github.com/ThanhDC-FSD/Job-Ops-Console/tree/showcase>

Professional Summary

- *Production-oriented engineer focused on **AI workflow orchestration**, reliable automation, and evidence-grounded outputs (RAG + validation) on CPU-friendly stacks.*
- *Designed and operationalized end-to-end pipelines: ingestion/normalization, retrieval + rerank, quality gates, deterministic fallbacks, and artifact rendering (TXT/DOCX/PDF) with auditable metadata.*
- *Strong delivery in enterprise environments: CI/CD automation, migration tooling, secure APIs, and resilient integrations with measurable efficiency gains.*

Work Experience

Senior Full Stack Developer | Bosch Global Software Vietnam

Mar 2022 - Present

- Orchestrated an HP ALM → GitHub migration and delivery automation pipeline (**Java**, Maven, RTC integrations, **Jenkins**), preserving history/metadata and reducing manual release steps by **~60%** while **improving** traceability.
- Automated CI/CD workflows (**Jenkins** Groovy, **Docker**, **docker-compose**, **Nginx**) with rollback patterns, cutting deployment turnaround time by **~30%** and reducing repeatable release errors.
- Implemented resilient data workflows for **PostgreSQL** (**SQLAlchemy**/Alembic), including daily automated backups (pg_dump + restore validation) to **improve** recovery confidence and operational reliability.
- Productionized secure API services (**Python FastAPI**, **SQLAlchemy**) with enterprise authentication (**Azure MSAL**, **JWT**, **cryptography**), enabling controlled access to internal systems.

- Operationalized real-time workflow feedback using WebSockets and structured event logging, reducing time-to-diagnose for runtime issues by ~25% during active development cycles.
- Built and hardened automation assistants: prompt template standardization (React + TypeScript + Zod), grammar enhancement (LanguageTool + spaCy), and tool integrations that improved consistency and reduced rework.
- Delivered RAG-enabled internal tooling (FastAPI, LangChain, Chroma) with practical retrieval + recommendation workflows to surface actionable knowledge and reduce time spent searching documentation.

Personal Project | Render CV Automation and MP4 Processing & Translation Platform

- Orchestrated an end-to-end AI automation platform for job applications: LinkedIn crawling, normalization, triage queues, CV/JD fit evaluation, and assisted apply workflows in a single operational dashboard.
- Implemented a production-style RAG pipeline on SQLite: embedding + hybrid retrieval (semantic/keyword/metadata), reranking, evidence mapping, and prompt contracts to keep outputs grounded and auditable.
- Added reliability gates and deterministic fallbacks (no-op rewrite detection, coverage deltas, constraint filters, mandatory cover-letter enforcement), improving stability on CPU-only hardware and reducing failed runs by ~40%.
- Automated document workflows to render tagged plain text into consistent DOCX/PDF artifacts (python-docx, docx2pdf), reducing manual formatting time by ~80% and improving consistency across versions.
- Contributed to a project with UI, MCP, and RAG capabilities using React, FastAPI, LangChain, and Chroma to support project management from existing knowledge, proactively surface delivery risks, and recommend practical mitigation actions, reducing issue-triage and context-gathering time by ~35%.
- Built a slide-generation project using Python, LLMs, Mermaid, and an Angular-based UI to transform prompts and structured inputs into presentation-ready slide outlines, diagrams, and editable content, reducing manual first-draft slide preparation time by ~70% and improving outline consistency across iterations.
- Built automation pipelines for media processing: subtitle extraction, multilingual translation, and speech-to-text (faster-

whisper, Transformers, Google Translate API), **improving** turnaround time for content localization by **~50%**.

- Delivered operational APIs and dashboards (**FastAPI**, **Uvicorn**, **Pydantic**, **React**, **Vite**) with analytics and stateful UX (**D3 Geo**, **TopoJSON**) to support high-throughput review of large job datasets.
- Implemented a pragmatic local CI/CD loop (scheduled deployments, sanitized publisher repo, local bare Git remote) to continuously ship updates while isolating runtime secrets.

Senior Java Developer and Oracle Middleware Engineer | GTEL ICT

Sep 2020 - Feb 2022

- Directed backend delivery for a national-scale system (**Spring Boot**, **Spring Security**), **improving** service robustness, access control, and operational stability.
- Implemented BI and **ETL** workflows (**Oracle OBIEE**, **ODI**), enabling reliable analytics pipelines and maintainable fact/dimension models for reporting.
- Optimized integrations with **SQL Server** using caching (EhCache) and schedulers, reducing latency and **improving** steady-state performance for batch workloads.

Java Developer | FPT Software

Jun 2019 - Sep 2020

- Implemented data ingestion automation for **Oracle/PostgreSQL** APIs, hardening large Excel imports with validation and error taxonomies to reduce operational incidents.
- Automated repeatable test/data workflows (Winium), **improving** throughput and reducing manual QA effort for regression tasks.

Java Developer (Contractor) | Tien Phong Bank

Feb 2019 - Jun 2019

- Automated a Data Exchange BOT (**Java** Swing, Selenium) for scheduling, reporting (PDF/TXT), and database synchronization, reducing manual operational steps by **~70%**.
- Implemented resilient ESB components (ESQL) to **improve** stability of card-service integrations and reduce recurring processing failures.

Java Developer | CODEGYM Vietnam

Aug 2018 - Feb 2019

- Delivered backend services (Spring Boot) and REST APIs with maintainable data models, improving delivery consistency across multiple training products.
- Implemented MySQL-backed persistence with unit tests and code review practices to improve correctness and reduce regressions.

Technical Skills

- **AI Workflow / RAG:** RAG pipelines, hybrid retrieval, reranking, evidence mapping, prompt contracts, validation gates, deterministic fallbacks, LangChain, Chroma, FAISS, sentence-transformers, qwen2.5 (offline gateway)
- **Workflow Automation:** Orchestration patterns, schedulers (APScheduler/cron), job queues, browser automation (Playwright), data normalization, fit scoring, integration reliability
- **Languages:** Java, Python, JavaScript (ES6+), TypeScript, SQL
- **Backend:** FastAPI, Starlette, SQLAlchemy, Pydantic, Uvicorn, Flask, Django, Spring Boot
- **Data / ETL:** pandas, numpy, openpyxl, XlsxWriter, ExcelJS, ETL pipelines, Oracle ODI/OBIEE
- **DevOps / Reliability:** Jenkins (Groovy), Docker, docker-compose, Nginx, Linux, rollback strategies, backup/restore validation
- **Databases:** PostgreSQL, SQLite, Oracle, SQL Server, MySQL
- **Frontend (Supportive):** React, Vite, HTML5, CSS3, Angular, D3 Geo, TopoJSON, world-atlas
- **Security:** Azure MSAL, JWT, passlib, cryptography, Spring Security
- **Others:** httpx, requests, Jinja2, python-docx, docx2pdf, python-pptx

Core Competencies

- AI Workflow Orchestration and Automation Engineering
- Production RAG Pipelines and Evidence-Grounded Generation
- Validation Workflows, Reliability Gates, and Deterministic Fallbacks
- CI/CD Automation, Migration Tooling, and Operational Hardening
- ETL/Data Workflows, Analytics, and High-Throughput Processing
- Secure API Design and Enterprise Authentication
- Cross-Functional Delivery, Ownership, and Collaboration